
Can a Knowledge-Graph Agent Close the Gap Between a Low-Cost and a Frontier-Class LLM? An Audited Upstream Petroleum Case Study

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Abstract

Small, low-cost language models are attractive agent backbones but are assumed to trail frontier models on tasks requiring exact identifiers, deterministic aggregation, and multi-hop reasoning. We test how close a knowledge-graph (KG) agent brings them. One ReAct pipeline – a structured graph-query tool, a deterministic aggregation tool, and a citation-traceability guardrail – runs a 50-question upstream-petroleum benchmark with a low-cost backbone (Gemini 3.1 Flash-Lite) and a frontier-class one (Claude Opus 4.8), three repeats each. Normalized exact match (EM) gives **0.898** vs. **0.889**; a symmetric hand audit of all 216 EM answers gives **0.926** vs. **0.954**, and the backbones differ on only 3 of 36 items, so this benchmark cannot separate them. The audit also shows that most strict-EM failures of both backbones are correct answers (82% and 81%), that grading rules derived from one backbone missed a format of the other, and that 7 of the 50 items are defective. An ablation ladder places the gain in the graph plus deterministic tools: graph retrieval without agency scores **0.278**, below Text-RAG at **0.583**. Since gold answers are computed from the graph the tools query, we read this as both backbones driving one toolchain to similar accuracy, not as a claim about model size.

1 Introduction

Small language models (SLMs) are attractive as agent backbones: cheaper, faster, and often free-tier-accessible, which matters for who can afford to deploy agentic tools at all. The standard assumption is that this comes at a capability cost on tasks needing exact identifiers, deterministic aggregation, and multi-hop reasoning – exactly the profile of upstream petroleum analytics, where analysts mix identification (*which well*), risk assessment (*is this pair a breakthrough risk*), recommendation (*what is the cheapest fix*), and causal explanation (*why*) over well topology, injector-producer connectivity, time-series sensor readings, alerts, and recommended interventions. We ask how much of that capability cost a better-structured pipeline can remove, rather than a bigger model.

Standard text-based RAG treats this domain as unstructured passages: it can retrieve topically relevant text, but it cannot compute rankings or aggregates, and it is prone to inventing plausible-looking but fake identifiers (e.g., intervention IDs) when asked to rank or recommend. Knowledge-graph RAG (KG-RAG) [1, 2, 3, 4] promises structure-aware retrieval and agentic tool use [5, 6] promises deterministic computation, but whether combining both lets a low-cost backbone do as well as a frontier one, and how such a system behaves when the graph is incomplete or noisy, remains relatively underexplored.

We present a knowledge-graph agent that combines (1) a structured graph-query tool, (2) a deterministic aggregation tool, and (3) a citation guardrail that rejects identifiers no tool returned, and evaluate

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it under two API backbones and against text-RAG and graph-retrieval-only baselines on a 50-question benchmark built from the public Volve field dataset, used here as a demanding testbed rather than as the paper’s primary claim. We study two research questions. **RQ1 (backbone dependence):** with the same KG agent, how close do a low-cost and a frontier-class backbone come, and which parts of the scaffold account for the result? **RQ2 (robustness, preliminary):** how does the agent’s accuracy change when the graph is incomplete or its connectivity weights are noisy? Our contributions are:

1. A cross-backbone comparison with a symmetric audit (RQ1): under one pipeline the two backbones come within one EM item per repeat of each other (audited EM 0.926 vs. 0.954), a difference this 36-item benchmark cannot resolve; we report similar accuracy, not parity (Section 5).
2. An instance-level audit of both backbones’ answers (RQ1) that separates model errors from grader artifacts and benchmark defects: most strict-EM failures of both backbones are correct answers, a normalized grader built from one backbone’s failures missed a format of the other, and seven benchmark items are defective.
3. An ablation ladder (RQ1) in which graph structure alone scores *lower* than text retrieval and the deterministic aggregation tool accounts for the largest single share of the full agent’s score.
4. A preliminary robustness probe (RQ2) on the low-cost backbone, with the perturbations, and the evidence they can and cannot reach, stated exactly.

Code, data, run logs, and audit verdicts are available at <https://github.com/TousifAhamed/kg-agent-low-cost-vs-frontier>.

2 Related Work

KG-RAG and graph-structured retrieval. GraphRAG [2] and related work [3] summarize community structure in a knowledge graph to answer global, corpus-level questions; Think-on-Graph [7] performs beam search over KG relations, showing that a strong graph can let a smaller LLM match a larger one, a hypothesis we test in a new domain. Pan et al. [4] survey the broader space of unifying LLMs and KGs. We build on classical RAG [1] (see Gao et al. [8] for a survey of the RAG design space) but replace passage retrieval with structured graph queries plus deterministic tools.

Agentic tool use. Chain-of-thought prompting [9] established that intermediate reasoning traces improve LLM task performance; ReAct [5] interleaves such traces with tool actions, and Toolformer [6] teaches models to invoke APIs. Qin et al. [10] and Wang et al. [11] survey the broader tool-use and autonomous-agent literature. Our agent follows the ReAct pattern but adds a domain-specific deterministic aggregation tool and a post-hoc citation guardrail, rather than relying on the LLM to compute rankings itself.

Ontologies and hallucination. We anchor entities to a minimal subset of the ISO 15926 process-industry data model [12], maintained by the POSC Caesar Association, within a knowledge graph [14] that carries a W3C PROV-O provenance profile [13], so every tool result carries a traceable node id. Ji et al. [15] survey hallucination in NLG; our citation guardrail is a lightweight, verifiable traceability check in the spirit of self-verification approaches such as Self-RAG [16].

Domain analytics. Capacitance-resistance modeling (CRM) [17, 18] and Chan’s water-cut/WOR diagnostic plots [19] are standard reservoir-engineering tools for inferring injector-producer connectivity and diagnosing coning/channeling; we encode their outputs as first-class KG edges and node properties rather than leaving them to unstructured retrieval.

3 Method

Knowledge graph. We build a property graph [20] over production and injection records from the public Volve field dataset (5 producers, 2 injectors, 2007–2016). Alerts, interventions, and derived diagnostic edges are not in the raw field release; they are generated once from the real series by fixed rules (first water cut $\geq 0.50 \rightarrow$ breakthrough alert, water cut $\geq 0.90 \rightarrow$ severe alert, $\log_{10} \text{WOR} \geq 0 \rightarrow$ high-WOR alert, one recommended intervention with sampled cost and uplift per breakthrough or

severe alert; Appendix A). This single instantiation is the KG seed for all experiments. The graph has 7 node types (Well, Reservoir, Sensor, MetricReading, Alert, Intervention, Field) and 6 relation types, each carrying a PROV wasDerivedFrom edge [13] (provenance coverage 1.0). All reported runs query an in-memory NetworkX copy of the graph; a Cypher export for Neo4j 5.26 exists but was not queried in these experiments.

Agent. An entity linker [21] seeds the first turn with mentioned KG node ids; a ReAct loop (max 5 tool rounds) then selects among `cypher_query` (despite its name, a structured tool: the model picks an operation – label/filter search, neighbours, k -hop subgraph, latest reading – and filters, rather than writing Cypher), `compute_metric` (deterministic count / sum / mean / min / max / rank, including derived ratios such as cost-per-barrel), and `semantic_search` (FAISS 1.9.0 [22] over the serialized-dashboard corpus of the Text-RAG baseline, as a fuzzy fallback). A citation guardrail checks *traceability*, not support: it accepts a cited id only if the id exists in the graph and was returned by a tool during that run (retrying up to twice), so it blocks fabricated identifiers but does not verify that a cited node supports the claim made about it. The low-cost backbone is `gemini-3.1-flash-lite` [23], accessed through the Gemini API free tier; it is Google’s most cost-efficient Gemini 3-series model and its parameter count is not published, so we call it “small” and “low-cost” in the sense of price tier only. The large backbone is `claude-opus-4-8` [24], a frontier-class commercial model at the time of the study. Both use the provider’s default decoding temperature and a 1024-token output cap, and the full agent was run three independent times on each. Large-backbone runs were made on 2026-07-03 and small-backbone runs on 2026-07-17–22, through floating API aliases without logged snapshot identifiers; the two-week gap is a confound we cannot remove, and exact re-execution against the same weights is not guaranteed.

Baselines. *S1 (LLM-only)*: no retrieval or tools. *S2 (Text-RAG)*: FAISS dense retrieval [25] ($k=10$) over serialized dashboard documents with the `all-MiniLM-L6-v2` Sentence-BERT [26] embedder (sentence-transformers 3.3.1). *S3 (KG-retrieval-only)*: k -hop subgraph serialization into a single LLM call, with no tools and no agent loop.

4 Experimental Setup

Benchmark. 50 questions (13 Identification, 12 Risk, 13 Recommendation, 12 Explanation), instantiated from templates by a script that computes each gold answer by traversal or aggregation over the KG; Appendix A lists all 50. The templates were drafted before any agent code existed, but were instantiated against the KG in the same week the agent’s tools were built, so tool design was not blind to the question set. We report EM over the 36 string- or number-answerable items, NDCG@5 [27] over the 2 ranking items, citation validity, and a 0–3 rubric over the 12 Explanation items, scored by an LLM judge [28] given the gold evidence nodes; the judge is the backbone being scored, so rubric scores are not comparable across backbones (Appendix D).

Grading. Strict EM applies a 2% numeric tolerance and ordered number matching for multi-number answers. Normalized EM additionally scans every number in a prose answer (strict takes only the first, which can pick the “5” out of “F-5”), normalizes prose dates, squashes underscores and hyphens, and discounts well ids echoed from the question (Appendix B). These four rules were written *after* auditing the small backbone’s failures, so they are tuned on one arm of the comparison, and a check that S1–S3 are unchanged by them cannot detect a rule that favors prose. We therefore also audit every EM answer of both backbones by hand against the question as worded (Section 5, Appendix C).

Repeats and perturbations. The full agent runs three times per backbone; the two ablations and the four perturbation conditions run once, on the small backbone only. Edge dropout at rate p removes each `INJECTS_INT0` edge and each `MetricReading` node independently with probability p ; weight noise at relative σ multiplies each `INJECTS_INT0` connectivity weight by $1 + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2)$, clipped at 0. Each condition is a single draw (seed 42), with gold answers fixed to the clean graph. Neither perturbation touches alert, intervention, or well nodes, and the `semantic_search` fallback indexes the unperturbed corpus.

5 Results

Similar accuracy, not parity (RQ1, Table 1). Under normalized EM the small backbone scores 0.898 and the large 0.889. But normalization lifts the small backbone by 15.7 points and the large

Table 1: Full agent, 36 EM items, 3 repeats per backbone (108 answers each). “Audited” judges every answer by hand against the question as worded, with corrected gold for defective items (Appendix C).

	Small (Flash-Lite)	Large (Opus 4.8)
Strict EM	0.741	0.852
Normalized EM (repeats 1/2/3)	0.898 (.917/.861/.917)	0.889 (.889/.889/.889)
Audited EM (repeats 1/2/3)	0.926 (.944/.889/.944)	0.954 (.972/.944/.944)
Strict-EM failures	28	16
artifact, fixed by normalization	17	4
artifact, missed by normalization	0	2
defective item, correct as worded	6	7
genuine error	5	3
Normalized credit, wrong on audit	3	2

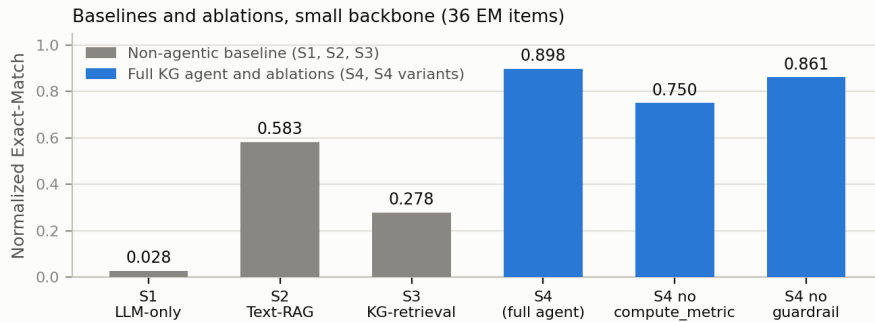


Figure 1: Normalized EM on the 36 EM items, small backbone. S4 is a 3-repeat mean; the two S4 ablations and the baselines are single runs. S2 and S3 score the same with the large backbone; S1 scores 0.056 there.

by 3.7, and its rules came from the small backbone’s failures, so we audited all 216 EM answers of both. Every answer the normalized grader newly accepts is correct for both backbones, so we found no credit for incidentally mentioned numbers; but the rules miss a format only the large backbone uses (“5/5 = 100%”), and the audit exposed two defective EM items: Q19’s gold value comes from a generator bug (the correct answer is “yes”), and Q25 asks how many producers “currently” exceed 90% water cut (one) while its gold counts those that ever did (two). On audit the scores are 0.926 (small) and 0.954 (large). The backbones’ item-level means differ on 3 of the 36 items, all in the large backbone’s favor (two-sided sign test $p = 0.25$; paired item bootstrap 95% CI for small minus large $[-0.065, 0.000]$; under normalized EM $[-0.019, +0.046]$). The benchmark therefore cannot separate the two backbones, and the audited point estimate favors the large one by one item per repeat. S2 and S3 score identically for both backbones (0.583 and 0.278); S1 scores 0.028 (small) and 0.056 (large), the extra credit being a non-answer to the defective Q19 that the boolean grader counts as “no”. Because the backbones also tie without the agent, these results show both reaching similar accuracy through the pipeline, not the pipeline closing a gap measured here.

What the scaffold contributes (RQ1). The ablations (Figure 1, small backbone) show graph structure alone is not enough: S3 (0.278) scores below S2 (0.583), since a serialized raw subgraph is a weaker signal than ranked dense passages for this question set. Removing `compute_metric` drops normalized EM from 0.898 to 0.750 (strict 0.667) and NDCG@5 from 1.000 to 0.659 – the agent still *finds* the right entities but can no longer *rank* them. Removing the guardrail costs less EM (0.861) and lowers citation validity from 1.00 to 0.98. These effects are partly built into the benchmark: gold answers to aggregation items are, by construction, what `compute_metric` returns over the same graph, and S1 scoring 1–2 of 36 shows the items are unanswerable without the tools. The ablation therefore says which tools this toolchain needs, not how graph and text retrieval compare in general. Rubric scores and NDCG@5 are in Appendix D.

Robustness, preliminary (RQ2, Figure 2). Normalized EM at 0 / 30 / 60% edge dropout is 0.898 / 0.694 / 0.778, and at relative weight noise $\sigma = 0 / 0.3 / 0.6$ it is 0.898 / 0.833 / 0.889; all four perturbed scores stay above the undegraded Text-RAG baseline (0.583). These numbers are weaker evidence

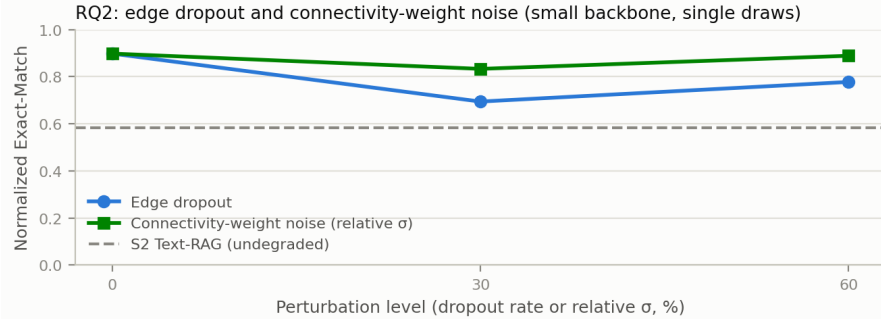


Figure 2: Normalized EM under edge dropout and connectivity-weight noise, small backbone, gold fixed to the clean graph. Each perturbed point is one draw; the 0% point is the 3-repeat mean. Dashed line: undegraded Text-RAG.

than they look. Of the 36 EM items, 20 rest only on alert, intervention, or well properties that neither perturbation alters, weight noise reaches the evidence of only 8, and the fallback search can still return clean values. On those 8 weight items, noise lowers EM from 0.833 to 0.625 ($\sigma=0.3$) and 0.750 ($\sigma=0.6$), while untouched items barely move; under 30% dropout, both touched (0.854 $\rightarrow$ 0.625) and untouched (0.933 $\rightarrow$ 0.750) items fall, which single draws cannot attribute to the perturbation rather than run-to-run variation. Neither curve is monotonic, and with one backbone the probe does not test whether similar accuracy holds under corruption.

6 Limitations and Future Work

Benchmark. The 50 items are generated from the same KG the agent queries, with gold defined as traversals or aggregations over it; we have no independently authored items (e.g., from field reports or engineers), so the results show that both backbones can drive this toolchain, not that the agent solves independently posed tasks. Only 36 items are EM-graded, which limits resolution to differences of a few points. Seven items are defective (Q16, Q19, Q21, Q25 among EM items; Q39, Q40, Q43 among rubric items; details in Appendix A); we flag them rather than patch them, and on the 32 non-defective EM items audited EM is 0.948 (small) vs. 0.969 (large). The alerts and interventions are rule-derived from one KG seed whose stability across regenerated seeds we have not tested. **Grading.** The normalized rules were fixed after the small-backbone audit and are not tested on held-out items; the audit is a single annotator’s judgment. The rubric judge is the scored backbone itself and is not validated against human raters. **Backbones and cost.** The two backbones are API models called through floating aliases two weeks apart; “small” refers to price tier, since Flash-Lite’s parameter count is unpublished, and we measured neither latency nor token cost, so the cost motivation in Section 1 is not tested here (Appendix E). **Robustness.** Single draws, one backbone, and perturbations that leave most evidence and the fallback corpus intact make RQ2 preliminary. All results are for one benchmark in one domain.

Broader impacts. A citation-guardrailed agent could make LLM-assisted field decision support more auditable than free-text RAG, since every cited entity is traceable to a KG node and tool call, and a low-cost backbone that performs close to a frontier-class one could lower deployment cost for smaller operators. But the agent is far from reliable (normalized EM 0.750 without the aggregation tool and as low as 0.694 under perturbation), traceable citations are not verified support, and our audits found seven defective benchmark items that automatic grading did not reveal; deploying such an agent in safety- or environmentally-critical decisions without human review risks automation bias, where a fluent, well-cited answer is trusted past its accuracy.

7 Conclusion

On this upstream petroleum benchmark, one knowledge-graph agent brings a low-cost and a frontier-class backbone to within one EM item per repeat of each other (audited 0.926 vs. 0.954), a gap the benchmark cannot resolve. The ablation ladder shows where that comes from: graph retrieval alone

reaches 0.278 and text retrieval 0.583, while the full agent reaches 0.898 and gives up 14.8 points without the deterministic aggregation tool. Because the questions are answerable only through that toolchain, the finding is that both backbones can drive it to similar accuracy, not that knowledge graphs make small models as capable as large ones. The most transferable lesson is methodological: auditing both backbones’ answers changed the sign of the headline difference and exposed seven defective items that automatic grading had hidden. Testing on independently authored questions, with repeated perturbations and measured cost, is the next step.

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Use of AI tools. An AI coding and writing assistant helped implement the pipeline, generate figures and tables from the stored results, draft the instance-level audit verdicts in Section 5, and draft and revise this manuscript. The author checked every audit verdict against the stored answers and the KG, and specified, checked, and approved all design choices, result numbers, and claims.

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A Benchmark and generation rules

Generation rules. Wells, connectivity, and production/injection series come from the Volve field release; CRM connectivity weights use a 45-day time constant for every injector-producer pair. Alerts are raised by fixed rules on the monthly readings: a `WATER_BREAKTHROUGH` alert (severity medium) at a producer’s first water cut ≥ 0.50 , a `WCT_SEVERE` alert (severity high) at every water cut ≥ 0.90 , and a `HIGH_WOR` alert at every $\log_{10}$ WOR ≥ 0 . Each breakthrough or severe alert receives one recommended intervention (action sampled from water shutoff, rate reduction, gas-lift optimization, polymer injection) with a lognormal cost per barrel (mean 12 USD) and an expected uplift. The script `src/eval/build_benchmark.py` then instantiates the 50 question templates and computes each gold answer from these tables or the graph.

All 50 items. Table 2 lists every item. “Evid.” gives the evidence an EM item’s gold rests on, which determines what the RQ2 perturbations can reach: **W** connectivity weights (edge dropout and weight noise), **E** connectivity edges only (edge dropout), **R** metric readings (edge dropout), **A** alert, intervention, or well properties (neither). Defective items are marked: (a) the question asks which pairs but the gold is a count; (b) the gold comes from a generator bug (`injectors_of` returns no injectors for F-1 C and F-15 D because of the space in their names; F-1 C is fed only by F-4 and F-15 D only by F-5, so the correct answer is true); (c) two wells tie on the same date and the gold names one; (d) the question says “currently” (answer 1, only F-14 at 0.9662) but the gold counts producers that ever crossed 0.90 (2); (e) the gold evidence is one specific alert while the question admits others, so a correct answer citing a different valid alert scores 0 – the large backbone traced `A_WCT_SEVERE_F-14_2016-07-01` for Q39 and `A_HIGH_WOR_F-12_2010-06-01` for Q43, each with the correct reading and value, and scored 0 in every repeat.

Table 2: The full benchmark. Defect marks (a)–(e) are explained above.

ID	Cat.	Question	Gold	Grader	Evid.
Q01	Id	Which injectors are connected to producer F-12 via the CRM connectivity fit?	F-4, F-5	EM	E
Q02	Id	What is the CRM connectivity weight from injector F-5 to producer F-11?	0.4368	EM	W
Q03	Id	What is the CRM time constant (tau, days) used for injector-producer pairs?	45.0	EM	E
Q04	Id	Which injector has the highest total outgoing connectivity weight across the field?	F-4	EM	W
Q05	Id	What is the dominant (highest-weight) supporting injector for producer F-11?	F-5	EM	W
Q06	Id	How many injector-producer pairs have a connectivity weight above 0.2?	4	EM	W
Q07	Id	What is the most recent $\log_{10}(\text{WOR})$ value for producer F-14?	1.4557	EM	R
Q08	Id	Which producers receive injection support from injector F-4?	F-1 C, F-11, F-12, F-14	EM	E
Q09	Id	What is the cumulative oil production (Np) for producer F-12 at the latest timestep?	4579609.55	EM	R
Q10	Id	What is the single strongest injector-producer connection in the field?	F-11, F-5	EM	W
Q11	Id	How many producer wells and how many injector wells are in the field?	5 producers, 2 injectors	EM	A
Q12	Id	What is the most recent water-cut fraction for producer F-12?	0.8835	EM	R
Q13	Id	On what date was producer F-14’s water-cut last measured?	2016-07-01	EM	R
Q14	Risk	Which producers have ever raised a <code>WCT_SEVERE</code> (water-cut $\geq 90\%$) alert?	F-12, F-14	EM	A
Q15	Risk	Which producers experienced a water-breakthrough alert?	F-1 C, F-11, F-12, F-14, F-15 D	EM	A
Q16 ^a	Risk	Which injector-producer pairs have connectivity weight above 0.3 (early-breakthrough risk)?	2	EM	W
Q17	Risk	How many <code>HIGH_WOR</code> alerts were raised across the field?	185	EM	A
Q18	Risk	Which producer has the greatest number of severe water-cut alerts?	F-14	EM	A
Q19 ^b	Risk	Do any producers depend on only a single injector (support single-point-of-failure)?	false	EM	E
Q20	Risk	How many total alerts were raised in the field?	234	EM	A

ID	Cat.	Question	Gold	Grader	Evid.
Q21 ^c	Risk	Which producer reached water breakthrough earliest (by alert date)?	F-12	EM	A
Q22	Risk	What fraction of producers ever crossed the 50% water-cut breakthrough threshold?	1.0	EM	A
Q23	Risk	Is producer F-12's connectivity strong enough (>0.2 from any injector) to risk injected-water arrival?	true	EM	W
Q24	Risk	Which producer carries the highest single connectivity weight (largest injected-water exposure)?	F-11	EM	W
Q25 ^d	Risk	How many distinct producers currently sit above the severe (90%) water-cut level?	2	EM	A
Q26	Rec	What is the single cheapest recommended intervention (by cost per barrel)?	I_WCT_SEVERE_F-14_2015-05-01	EM	A
Q27	Rec	Which intervention action type is recommended most frequently?	gas_lift_opt	EM	A
Q28	Rec	How many recommended interventions are there in total?	49	EM	A
Q29	Rec	Which intervention has the highest expected oil uplift (bbl)?	I_WATER_BREAKTHROUGH_F-15D_2015-12-01	EM	A
Q30	Rec	What is the mean estimated cost-per-barrel across all interventions?	12.65	EM	A
Q31	Rec	Which intervention offers the best expected-uplift-per-cost ratio?	I_WCT_SEVERE_F-14_2016-02-01	EM	A
Q32	Rec	How many interventions are recommended for producer F-14?	25	EM	A
Q33	Rec	Which producers have at least one recommended intervention?	F-1 C, F-11, F-12, F-14, F-15 D	EM	A
Q34	Rec	Rank the 5 cheapest interventions by cost-per-barrel (ascending).	1. I_WCT_SEVERE_F-12_2013-07-01 2. I_WCT_SEVERE_F-12_2014-02-01 3. I_WCT_SEVERE_F-12_2014-06-01 4. I_WCT_SEVERE_F-14_2015-05-01 5. I_WCT_SEVERE_F-14_2016-02-01	NDCG	–
Q35	Rec	What action is recommended for the cheapest intervention?	water_shutoff	EM	A
Q36	Rec	How many interventions use the water_shutoff action?	10	EM	A
Q37	Rec	For producer F-14, which recommended intervention has the lowest cost per barrel?	I_WCT_SEVERE_F-14_2015-05-01	EM	A
Q38	Rec	Rank the top 5 interventions by expected uplift (descending).	1. I_WATER_BREAKTHROUGH_F-15D_2015-12-01 2. I_WCT_SEVERE_F-12_2013-06-01 3. I_WCT_SEVERE_F-12_2013-11-01 4. I_WCT_SEVERE_F-14_2015-12-01 5. I_WCT_SEVERE_F-14_2016-02-01	NDCG	–
Q39 ^e	Expl	Explain why producer F-14 raised a severe water-cut alert: trace alert → triggering reading → recommended intervention.	rubric; evidence A_WATER_BREAKTHROUGH_F-14_2010-06-01	Rubric	–
Q40 ^c	Expl	Trace the earliest water-breakthrough alert: which reading triggered it, on what date, and what intervention was recommended?	rubric; evidence A_WATER_BREAKTHROUGH_F-12_2010-06-01, R_water_cut_F-12_2010-06-01	Rubric	–
Q41	Expl	Why is the F-5 → F-11 pair flagged as an early-breakthrough risk? Explain using its connectivity weight.	rubric; evidence C_F-5_F-11	Rubric	–

ID	Cat.	Question	Gold	Grader	Evid.
Q42	Expl	Explain the provenance chain of the cheapest recommended intervention back to its source record.	rubric; evidence I_WCT_SEVERE_ F-14_2015-05-01	Rubric	–
Q43 ^e	Expl	What evidence supports a HIGH_WOR alert (which reading and threshold)?	rubric; evidence A_HIGH_WOR_ F-1C_2014-10-01	Rubric	–
Q44	Expl	Explain the difference in water-cut trajectory between producers F-12 and F-15 D.	rubric; evidence W_F-12, W_ F-15D	Rubric	–
Q45	Expl	Why does producer F-11 receive more support from injector F-5 than from F-4? Explain using the weights.	rubric; evidence C_F-5_F-11, C_ F-4_F-11	Rubric	–
Q46	Expl	Explain, end to end, the risk-to-recommendation chain for the field’s highest-risk producer.	rubric; evidence 15/9-F-14	Rubric	–
Q47	Expl	Explain why every KG node carries a source_ record_id and why that matters for citation validation.	rubric	Rubric	–
Q48	Expl	What causes the recommended water_shutoff interventions, in terms of the water diagnostics?	rubric; evidence I_WCT_SEVERE_ F-12_2013-03-01	Rubric	–
Q49	Expl	Describe the connectivity structure of injector F-4: which producers it supports and the relative weights.	rubric; evidence W_F-4	Rubric	–
Q50	Expl	Summarize how an alert, its triggering reading, and its recommended intervention link together in the ontology.	rubric	Rubric	–

B Normalized EM grader

The normalized grader differs from the strict grader in exactly the four ways below. Gold answers are identical under both; only the parsing of the model’s free text differs, and both graders already apply a 2% numeric tolerance and ordered-subsequence matching for multi-number answers. The rules were written after the small-backbone audit of 2026-07-19 and first applied on 2026-07-20; the large backbone’s answers (2026-07-03) were not used to design them.

1. *Numeric-scalar extraction.* Strict grading extracts only the *first* number in the answer, which can pick a digit out of a well id (the “5” in “F-5”). Normalized grading accepts if *any* number in the text is within tolerance of gold.
2. *Date normalization.* Prose dates such as “July 1, 2016” are rewritten to YYYY-MM-DD before number matching.
3. *Punctuation squashing for single-token strings.* Underscores and hyphens are removed before substring matching, so “water shutoff” matches `water_shutoff`.
4. *Question-echo discounting for entity-list answers.* Well ids that also appear in the question are removed before comparing the predicted and gold sets.

Rule 1 could in principle credit an answer that mentions the gold value incidentally. In the audit (Appendix C) every one of the 21 answers that normalized EM accepts and strict EM rejects (17 small, 4 large) is a correct answer, so this did not occur in these runs. The rules do miss two formats: a fraction written as a percentage (“5/5 = 100%” for gold 1.0; two large-backbone answers), and the boolean grader counts a non-answer (“cannot determine”) as “no” (one S1 answer).

C Symmetric audit

Every EM answer of the full agent, 36 items $\times$ 3 repeats $\times$ 2 backbones, was judged against the question as worded, starting from the normalized verdict and overriding it only where the stored answer and the KG show the grader is wrong; for the defective EM items the judgment uses the corrected gold (Q16: the two pairs F-4 $\rightarrow$ F-11 and F-5 $\rightarrow$ F-11; Q19: true; Q21: either tied well; Q25: one). Table 3 lists every item whose verdicts are not uniformly correct; the other 23 items are answered correctly by both graders in all six runs. The paired comparison treats the 36 items as the unit: each backbone’s item score is its mean over three repeats, and the 95% interval is a percentile bootstrap over items (10,000 resamples). Because only 3 items differ on audit, all favoring the large backbone, the interval’s upper end is exactly 0 and the sign test is the more interpretable summary ($p = 0.25$). The verdicts, with a reason for each, and the script that

computes every number in Table 1 are in the code release (results/audit/ and src/eval/audit_both.py at <https://github.com/TousifAhamed/kg-agent-low-cost-vs-frontier>).

Table 3: Item-level audit. Each backbone shows three groups (repeats 1–3) of three marks: strict / normalized / audited.

Item	Small	Large	Reason
Q01	×✓✓ ✓✓✓ ×✓✓	✓✓✓ ✓✓✓ ✓✓✓	question-echo artifact
Q02	×✓✓ ×✓✓ ×✓✓	✓✓✓ ✓✓✓ ✓✓✓	first-number artifact
Q04	✓✓✓ ××× ✓✓✓	✓✓✓ ✓✓✓ ✓✓✓	genuine aggregation error (small r2)
Q07	×✓✓ ×✓✓ ×✓✓	✓✓✓ ✓✓✓ ✓✓✓	first-number artifact
Q09	×✓✓ ×✓✓ ×✓✓	×✓✓ ×✓✓ ×✓✓	first-number artifact
Q12	×✓✓ ×✓✓ ×✓✓	✓✓✓ ✓✓✓ ✓✓✓	first-number artifact
Q13	×✓✓ ✓✓✓ ✓✓✓	✓✓✓ ✓✓✓ ✓✓✓	prose-date artifact
Q16	××✓ ××✓ ××✓	××✓ ××✓ ××✓	gold defect (a)
Q19	××✓ ××✓ ××✓	××✓ ××✓ ××✓	gold defect (b)
Q22	✓✓✓ ××× ✓✓✓	×✓✓ ××✓ ××✓	first-number artifact (large r1); 5/5 = 100% missed by normalized grader (large r2, r3); genuine 4-of-5 count (small r2)
Q25	✓×× ✓✓× ✓✓×	××✓ ✓✓× ✓✓×	question defect (d)
Q33	××× ××× ×××	××× ××× ×××	genuine: omits F-15 D (both backbones)
Q35	✓✓✓ ×✓✓ ×✓✓	✓✓✓ ✓✓✓ ✓✓✓	underscore artifact

D Explanation rubric and ranking results

Each Explanation answer is scored 0–3 by an LLM judge that sees the question, the gold evidence node ids, and the answer, with the rubric “0 = wrong or unsupported; 1 = partially correct, key steps/evidence missing; 2 = correct reasoning with most evidence; 3 = fully correct end-to-end trace citing the right evidence nodes” and a system prompt rewarding correct alert → reading → intervention chains, correct use of connectivity weights, and reference to the gold nodes. The judge is the same backbone as the agent being scored (Flash-Lite judged Flash-Lite, Opus judged Opus), so the two columns of Table 4 use different judges and should not be compared directly. Q39 and Q43 are defective in a way that penalizes correct answers (mark (e), Appendix A); both backbones score 0 on Q39, and the large backbone’s 0 on Q43 in every repeat (against the small backbone’s 3) accounts for the entire difference in summed scores (79 vs. 70). NDCG@5 on the two ranking items is 1.000 for both backbones in every repeat.

Table 4: Rubric scores (0–3) per Explanation item, repeats 1 / 2 / 3, each backbone judged by itself.

Item	Small (Flash-Lite)	Large (Opus 4.8)
Q39	0 / 0 / 0	0 / 0 / 0
Q40	3 / 3 / 3	2 / 2 / 3
Q41	3 / 3 / 2	2 / 2 / 1
Q42	3 / 3 / 3	3 / 3 / 3
Q43	3 / 3 / 3	0 / 0 / 0
Q44	3 / 3 / 3	2 / 3 / 3
Q45	1 / 2 / 3	2 / 2 / 2
Q46	0 / 3 / 0	3 / 0 / 3
Q47	2 / 3 / 2	3 / 3 / 2
Q48	1 / 1 / 1	3 / 3 / 2
Q49	3 / 3 / 3	3 / 3 / 3
Q50	3 / 1 / 3	1 / 1 / 2
Mean	2.19	1.94

E Compute resources

All experiments ran on a single consumer laptop with no GPU: an 11th-generation Intel Core i7-1185G7 (4 cores / 8 threads), 32 GB RAM, Windows 11 Pro (10.0.26200), Python 3.11.15, PyTorch 2.5.1+cpu (CPU build; CUDA unavailable), sentence-transformers 3.3.1, faiss-cpu 1.9.0, and NetworkX 3.4.2 for the in-memory graph that every reported run queries (a Cypher export for Neo4j 5.26 was built but not used).

On-disk footprint is small: 2.3 MB raw Volve input, 203 KB generated tables, 1.6 MB materialized graph artifacts, 16 KB benchmark.

Local, non-API pipeline stages measured on that machine: knowledge-graph construction 3.5 s (deterministic – re-running reproduces byte-identical artifacts), Text-RAG corpus serialization 1.1 s for 304 documents, embedding model load 7.3 s, and embedding of the full corpus 5.3 s. The only substantial cost is API latency, not local compute.

Per-question LLM usage is bounded by the agent’s budget (max 5 tool rounds, max 2 citation retries). Measured from the stored run logs, the full agent issues a mean of 3.06 tool calls per question on the small backbone (max 9) and 2.41 on the large backbone (max 12), with one additional LLM call to produce the final answer; the two ablations average 2.94 and the perturbation conditions 3.37. The observed full 50-question runs required approximately 200 LLM calls – a typical run volume, not the theoretical maximum the budget allows (50 questions $\times$ up to 5 tool rounds plus up to 2 citation retries each).

We did not instrument per-run wall-clock time or token counts during the original experiments, so we cannot report measured runtime or token cost for those runs, and we do not estimate them retrospectively. Because cost and latency motivate the use of a low-cost backbone, measuring both is the most direct gap to close in future runs.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

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Justification: The Introduction states two research questions (RQ1 backbone dependence, RQ2 preliminary robustness) and four contributions mapped onto them. The abstract and Introduction state the cross-backbone result as similar accuracy that this benchmark cannot resolve, not parity; report both automatic (0.898 vs. 0.889) and audited (0.926 vs. 0.954) scores; and qualify all findings as specific to one 50-question benchmark in one domain whose gold answers are computed from the graph the agent queries. Every headline number in the abstract appears in Section 5 or follows from Table 1.

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Answer: [\[Yes\]](#)

Justification: The LLM is the core reasoning engine of the agent under study (Section 3), under both a low-cost and a frontier-class backbone compared in Section 5, and also serves as the rubric judge for Explanation items (Section 4, Appendix D). An AI assistant was also used to help implement the pipeline, draft the audit verdicts, and draft and revise the manuscript, as disclosed in the “Use of AI tools” paragraph; the author checked every audit verdict.

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